

Bangladesh's Landslide Prediction and Risk Management Database

1. Designing (Entity Relationship) ER Diagram

Steps of Drawing ERD

1. Identify the Entities Required
2. Identify the Attributes and Primary Key for each Entity
3. Identify the Relationship Needed
4. Identify the Cardinality Ratio and Participation
5. Draw the Diagram

A Landslide Prediction and Risk Management System focuses on tracking landslide occurrences, geological risk factors, and mitigation efforts in Bangladesh. This database system provides critical insights for disaster management, environmental protection, and governmental policy-making. Below is the introduction and implementation outline using MySQL.

Step-1: Identify the Entities Required

1. **Location**
2. **Landslide_Event**
3. **Risk_Assessment**
4. **Mitigation_Measure**
5. **Government_Agency**

Step-2: Identify the Attributes and Primary Key for each Entity

1. **Location** (Location_ID, District, Latitude, Longitude, Elevation, Soil_Type, Last_Assessed)
2. **Landslide_Event** (Event_ID, Location_ID, Event_Date, Casualties, Rainfall, Description)
3. **Risk_Assessment** (Assessment_ID, Location_ID, Assessment_Date, Risk_Level, Recommendations)
4. **Mitigation_Measure** (Measure_ID, Location_ID, Agency_ID, Measure_Name, Description, Implementation_Date)
5. **Government_Agency** (Agency_ID, Agency_Name, Contact_Person, Phone, Email)

Step-3: Identify the Relationship Needed

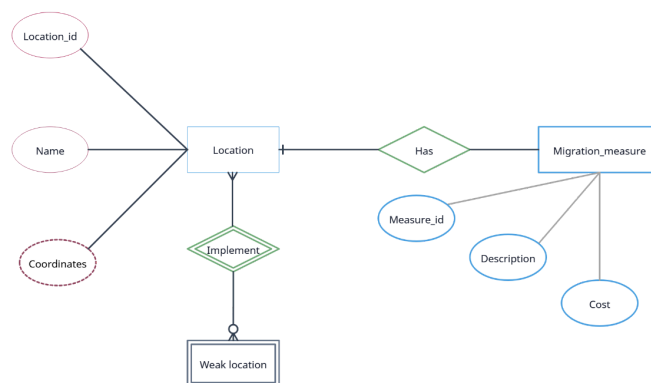
- Location → Has → Landslide_Event (One-to-Many)

- Location → Has → Risk_Assessment (One-to-Many)
- Location → Has → Mitigation_Measure (One-to-Many)
- Government_Agency → Implements → Mitigation_Measure (One-to-Many)
- Mitigation_Measure → Implemented_By → Government_Agency (Many-to-One)

Step-4: Identify the Cardinality Ratio and Participation

1. **Location → Has → Landslide_Event (One-to-Many)**
 - Every Location may have multiple Landslide Events, but each event occurs at one Location.
2. **Location → Has → Risk_Assessment (One-to-Many)**
 - Each Location may have multiple Risk Assessments.
3. **Location → Has → Mitigation_Measure (One-to-Many)**
 - Each Location may have multiple Mitigation Measures.
4. **Government_Agency → Implements → Mitigation_Measure (One-to-Many)**
 - A Government Agency may implement multiple Mitigation Measures.
5. **Mitigation_Measure → Implemented_By → Government_Agency (Many-to-One)**
 - A Mitigation Measure is implemented by one Government Agency.

Step-5: Draw the ER Diagram



2. Reduction to Database Schema

Table: **Location**

CREATE TABLE Location (

Location_ID INT PRIMARY KEY AUTO_INCREMENT,

District VARCHAR(50) NOT NULL,
Latitude DECIMAL(9,6) NOT NULL,
Longitude DECIMAL(9,6) NOT NULL,
Elevation DECIMAL(6,2) NOT NULL,
Soil_Type VARCHAR(20) NOT NULL,
Last_Assessed DATE
);

Table: Landslide_Event

```
CREATE TABLE Landslide_Event (  
    Event_ID INT PRIMARY KEY AUTO_INCREMENT,  
    Location_ID INT,  
    Event_Date DATE NOT NULL,  
    Casualties INT,  
    Rainfall DECIMAL(6,2),  
    Description TEXT,  
    FOREIGN KEY (Location_ID) REFERENCES Location(Location_ID)  
);
```

Table: Risk_Assessment

```
CREATE TABLE Risk_Assessment (  
    Assessment_ID INT PRIMARY KEY AUTO_INCREMENT,  
    Location_ID INT,  
    Assessment_Date DATE NOT NULL,  
    Risk_Level ENUM('Low', 'Medium', 'High') NOT NULL,  
    Recommendations TEXT,  
    FOREIGN KEY (Location_ID) REFERENCES Location(Location_ID)
```

);

Table: Government_Agency

```
CREATE TABLE Government_Agency (  
    Agency_ID INT PRIMARY KEY AUTO_INCREMENT,  
    Agency_Name VARCHAR(100) NOT NULL,  
    Contact_Person VARCHAR(100) NOT NULL,  
    Phone VARCHAR(20) NOT NULL,  
    Email VARCHAR(100) NOT NULL  
);
```

Table: Mitigation_Measure

```
CREATE TABLE Mitigation_Measure (  
    Measure_ID INT PRIMARY KEY AUTO_INCREMENT,  
    Location_ID INT,  
    Agency_ID INT,  
    Measure_Name VARCHAR(100) NOT NULL,  
    Description TEXT,  
    Implementation_Date DATE,  
    FOREIGN KEY (Location_ID) REFERENCES Location(Location_ID),  
    FOREIGN KEY (Agency_ID) REFERENCES Government_Agency(Agency_ID)  
);
```

3. Implementing the Database in MySQL

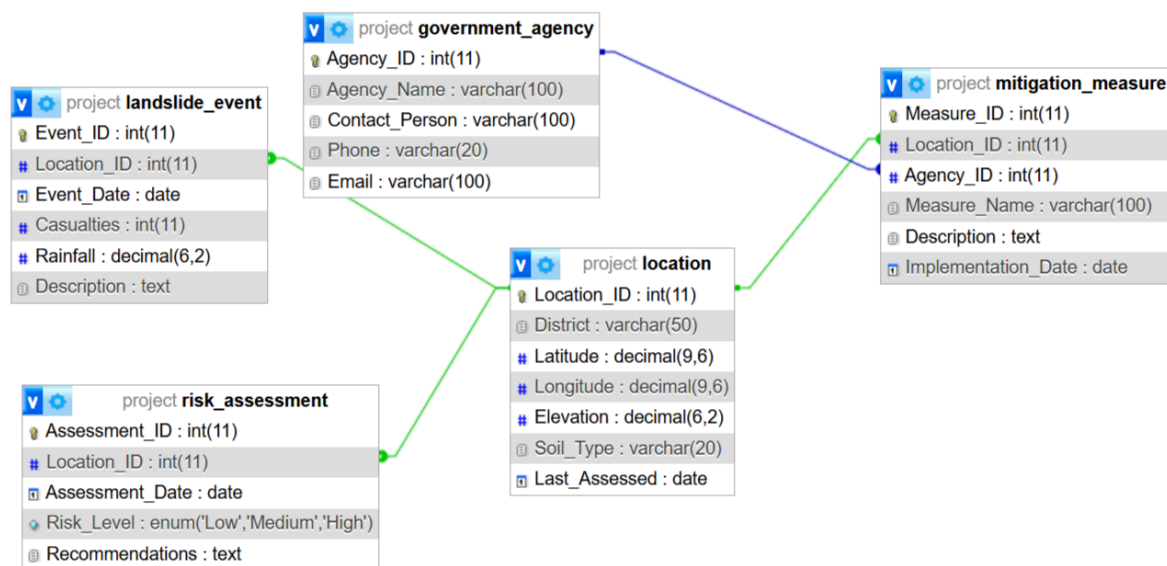
Sample Data:

```
INSERT INTO Location (District, Latitude, Longitude, Elevation, Soil_Type, Last_Assessed)  
VALUES
```

('Chittagong', 23.8103, 90.4125, 15.2, 'Clay', 2025-01-01');

INSERT INTO Landslide_Event (Event_id, Location_ID, Event_Date, Casualties, Rainfall, Description) VALUES

(1, 1, '2025-02-10', 5, 120.50, 'Severe landslide due to heavy rainfall');



Location Table

location_id	District	Latitude	Longitude	Elevation	Soil_type	Last_assessed
1	Chittagong	23.8103	90.4125	15.2	Clay	2025-01-01

Landslide_Event Table

Event_i d	Location_i d	Event_date	Casualtie s	Rainfal l	Description
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1	1	2025-02-10	5	120.50	Severe landslide due to heavy rainfall
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4. Query

1. Get all landslide occurrences in a specific region:

```
SELECT L.Event_Date, L.Casualties, L.Rainfall, LOC.District
FROM Landslide_Event L
JOIN Location LOC ON L.Location_ID = LOC.Location_ID
WHERE LOC.District = 'Chittagong';
```

2. Find high-risk areas with a probability greater than 70%:

```
SELECT LOC.District, RA.Risk_Level, RA.Recommendations
FROM Risk_Assessment RA
JOIN Location LOC ON RA.Location_ID = LOC.Location_ID
WHERE RA.Risk_Level = 'High';
```

5. Conclusion

This database provides a structured way to manage landslide prediction and risk data in Bangladesh. It enables efficient data storage, analysis, and retrieval for researchers and disaster management authorities. The system can be further enhanced by integrating GIS for real-time monitoring and AI-driven predictive analytics.

6. Future Scope

- Integration with GIS for real-time landslide tracking.
- Use of AI/ML models for better prediction.
- Development of a web-based dashboard for easy accessibility.

7. References

- Bangladesh Meteorological Department (BMD)
- Disaster Management Bureau (DMB)

- Research papers on landslide risk assessment in Bangladesh